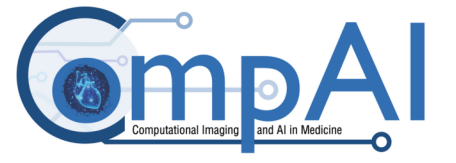


ConVIRT vs. MedCLIP - A Comparison



Revealing the role of balancing datasets and the influence of backbone architectures



Sandra Dening, Caterina Galatà

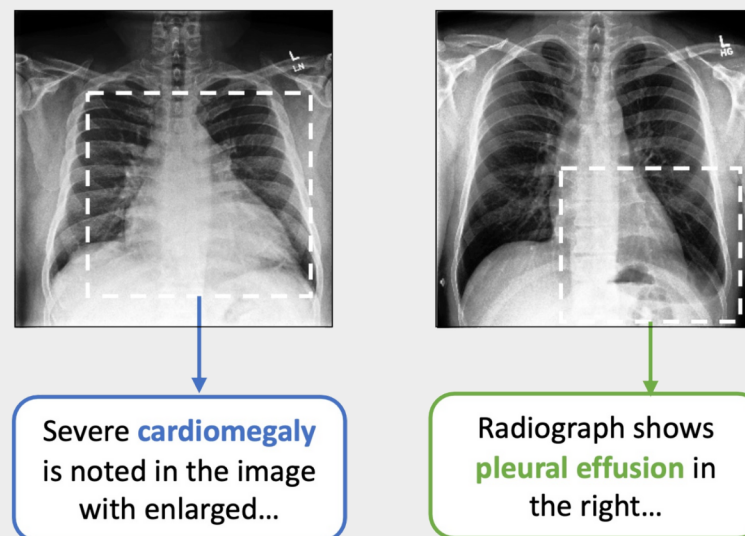
CHALLENGES IN MEDICAL IMAGING

- Diagnosis depends on **subtle visual cues**
- Overall **similarity of images** outside of the lesion region
- Extreme **scarcity of annotated data**
- **Generalization** to unseen data, diseases, patients, ...
- **Imbalance in data distribution**, as "normality" represents healthy cases, leading to a significant number of false negatives and low recall

VISION - LANGUAGE MODELING

ConVIRT^[1]

Learning visual representations by exploiting the naturally occurring **pairing of images and textual data**

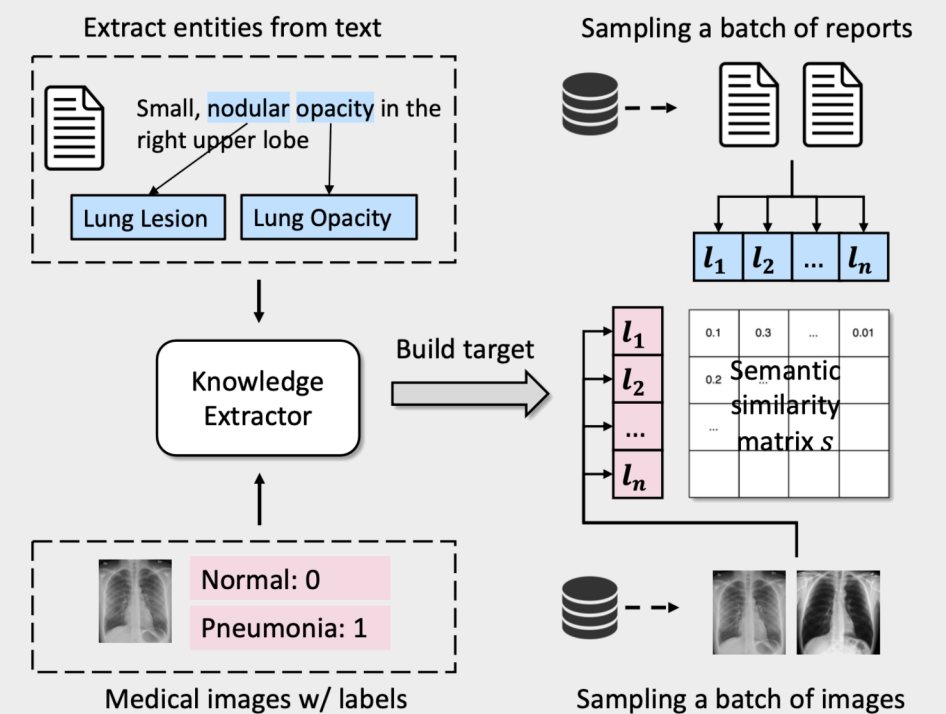


Using **contrastive learning** to learn positive and negative pairs

Differentiate samples belonging to different patients

MedCLIP^[2]

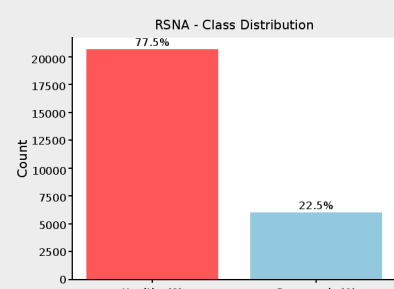
Leverage external medical knowledge to build the **knowledge-driven semantic similarity** using UMLS



Differentiate samples via their semantic similarity

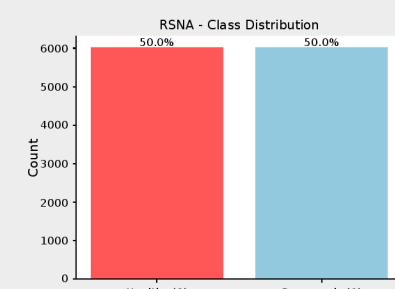
EXPERIMENTS AND RESULTS

Modeling class imbalance in the RSNA dataset - linear classification (1 FC layer), RSNA



UNBALANCED

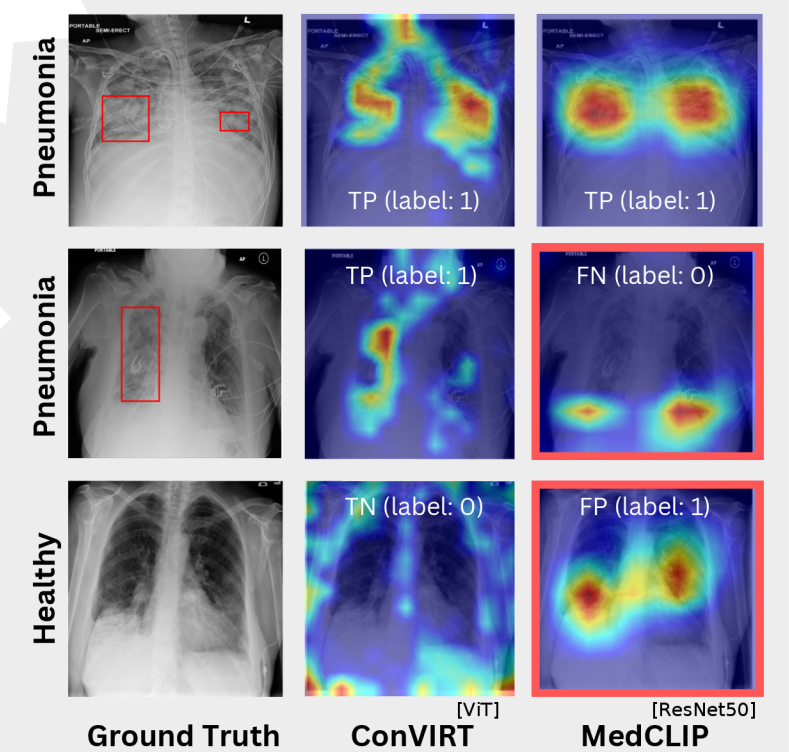
Undersampling



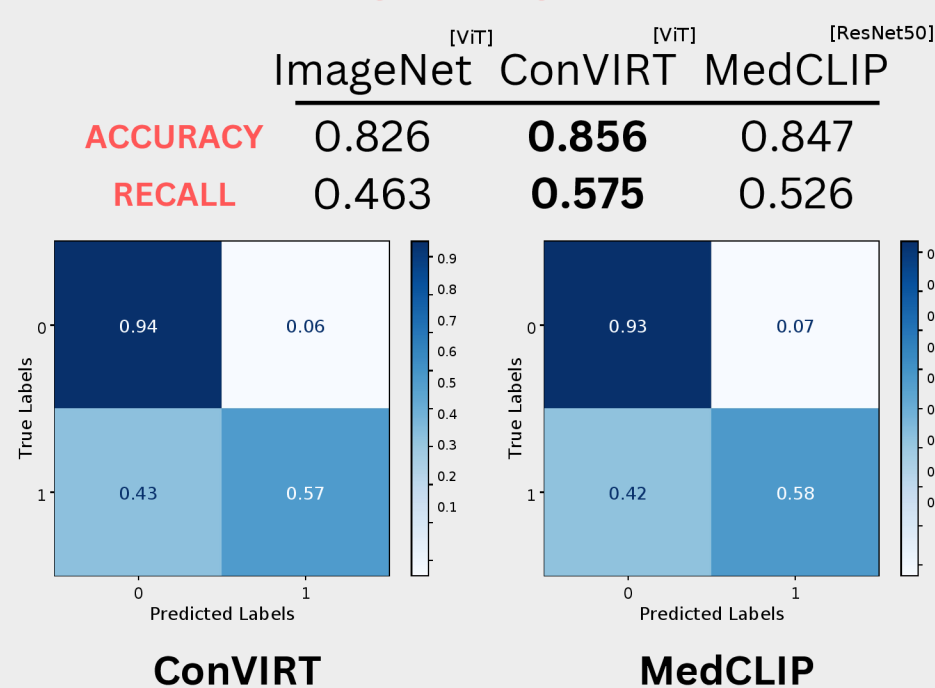
BALANCED

ConVIRT with ViT backbone performs comparably, diminishing the apparent advantages of MedCLIP

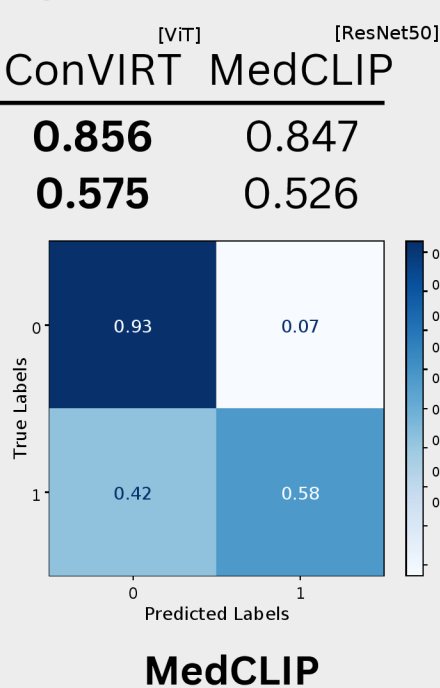
Saliency Maps of Predictions



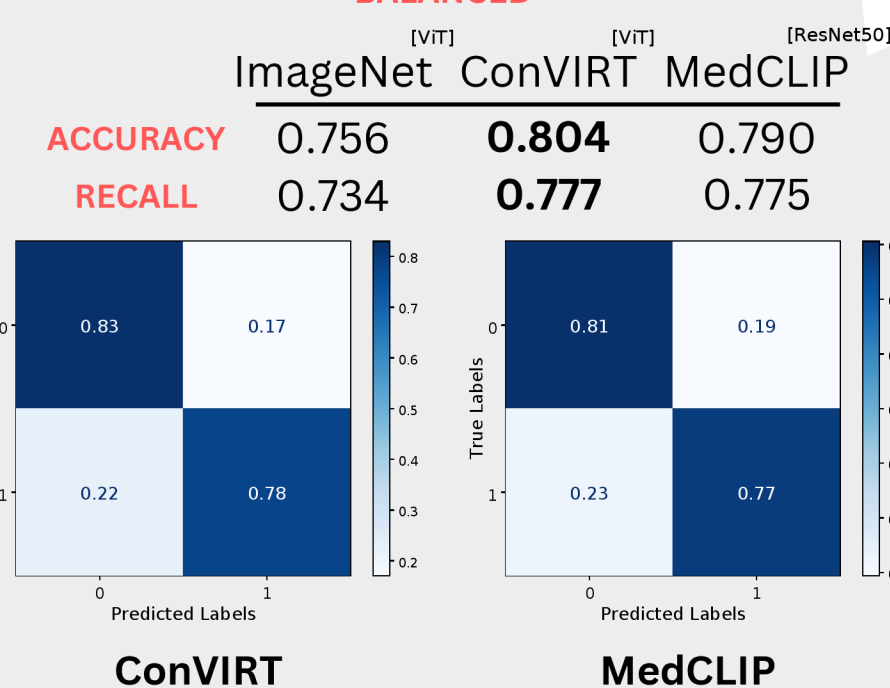
WHY?



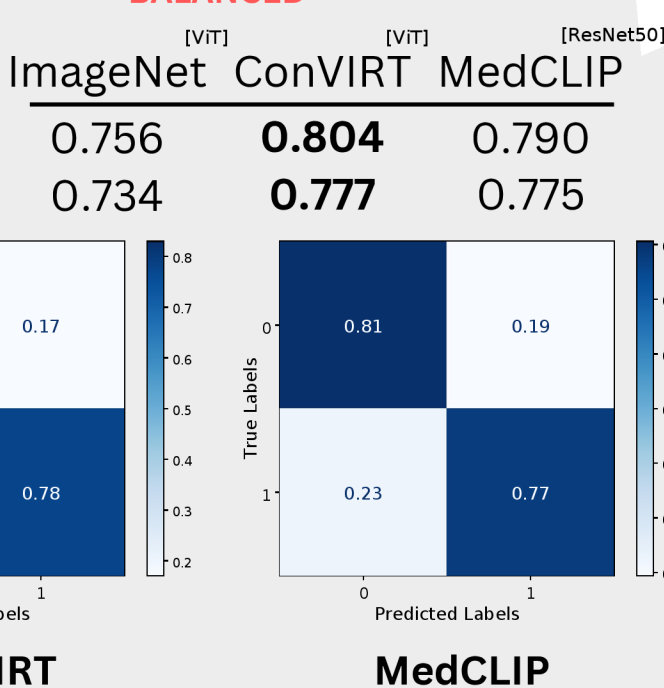
ConVIRT



MedCLIP

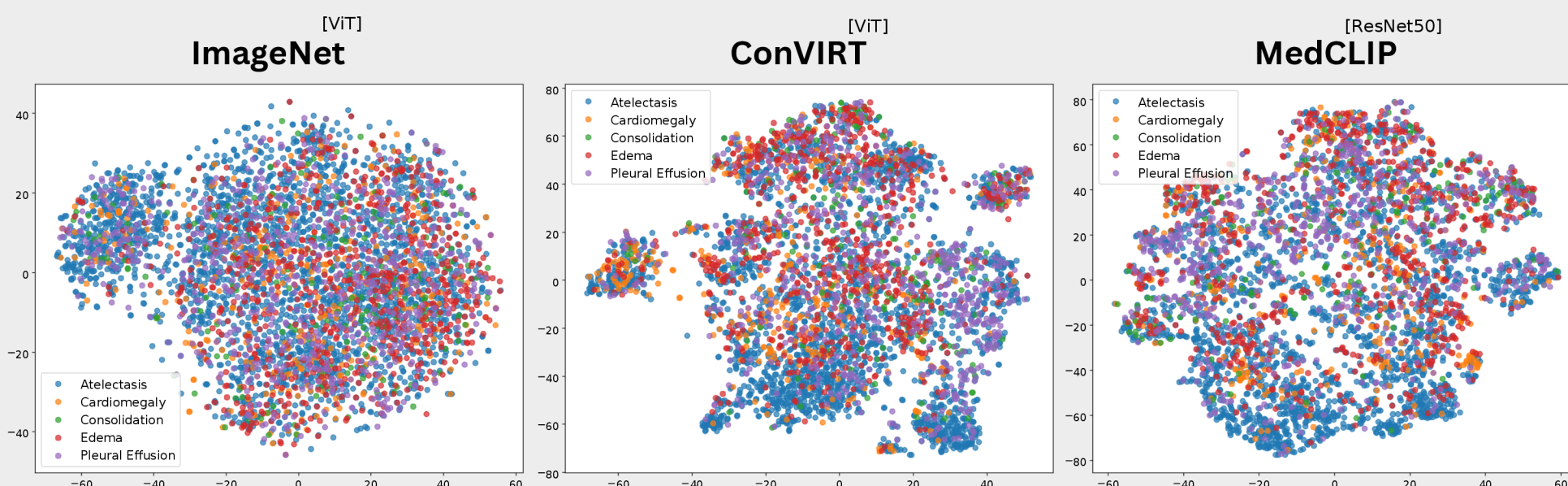


ConVIRT



MedCLIP

Image embeddings comparison - linear classification (1 FC layer), CheXpert 5x200



The t-SNE plots are rather homogeneous, due to overall similarity of X-ray images. Nonetheless, ConVIRT and MedCLIP still detect clusters by the lesion type.

Code repo:

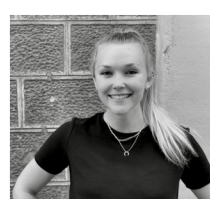


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References:

- [1] Zhang, Y., Jiang, H., Miura, Y., Manning, C. D., & Langlotz, C. P. (2020, October 2). Contrastive Learning of Medical Visual Representations from Paired Images and Text. *arXiv.org*. <https://arxiv.org/abs/2010.00747>
- [2] Wang, Z., Wu, Z., Agarwal, D., & Sun, J. (2022, October 18). MedCLIP: Contrastive Learning from Unpaired Medical Images and Text. *arXiv.org*. <https://arxiv.org/abs/2210.10163>